Change-Aware Cascaded Dual-Decoder Network for Remote Sensing Image Change Detection

Feng Yang, Yifeng Yuan, Anyong Qin, Yue Zhao, Tiecheng Song, *Member, IEEE*, and Chenqiang Gao

***Abstract*— Change detection aims to detect changes of objects or scenes in remote sensing images, which is critical for observing the Earth's surface. However, due to the insufficient correlation and aggregation of bitemporal features, the existing deep learning methods are still impacted by varied imaging conditions and complicated boundaries of ground objects in high-resolution remote sensing images. To tackle these challenges, we propose a change-aware cascaded dual-decoder network (CACD2Net), which integrates bitemporal features at different levels to facilitate learning change maps from coarse to fine, thus empowering the network to effectively identify changes and refine pixelwise boundaries in a progressive manner. Within the cascaded dual-decoder architecture, the change location decoder utilizes high-level features to generate a coarse change map, which approximates changes' localization, while the mask refinement decoder further leverages low-level features to create a texture-aware map that captures more texture and structural information about the change regions. By using the coarse change map as guidance and directing the texture-aware map to focus on the details of changes, the boundaries can be gradually refined, ultimately resulting in an accurate change detection mask. We test our model on the season-varying change detection (SVCD) dataset and the Sun Yat-sen University change detection (SYSU-CD) dataset, and the experimental results show that our model surpasses other state-of-the-art change detection methods. Our codes will be available at https://github.com/Moonquakes0/CACD2Net.**

***Index Terms*— Attention mechanism, cascaded dual decoder, change detection, difference enhancement, high-resolution remote sensing images.**

I. Introduction

CHANGE detection is a remote sensing image analysis technique for identifying and detecting changes of objects or scenes, which has been widely used in resource management [1], environmental monitoring [2], urban planning [3], and so on. High-resolution remote sensing images provide more abundant yet complicated content of the ground surface, introducing noise and interference (e.g., the varied imaging conditions and irregular geometrical structures of geospatial objects) into change detection. How to effectively extract discriminative information in high-resolution remote sensing images is very crucial for change detection.

Before the emergence of deep learning methods, algebraic-based approaches were commonly employed for change detection, such as image differencing [4], image rationing [5], change vector analysis [6], and image regression [7]. However, traditional methods are sensitive to noise, which makes it difficult to detect changes in high-resolution remote sensing images. In recent years, deep learning technology [8] has experienced rapid development. Autoencoder (AE) [9], [10], deep belief network (DBN) [11], [12], recurrent neural networks (RNNs) [13], [14], convolutional neural networks (CNNs) [15], [16], [17], and Transformer [18] are commonly used in change detection, and the use of deep neural networks offers discriminative capabilities and effective feature representation. Although these models show better performance than traditional methods, they still suffer from unsatisfactory results due to the varied imaging conditions and complex boundaries of ground objects.

The variations in imaging conditions, such as illumination and seasons, cause significant deviations for the objects in the bitemporal images. As shown in Fig. 1, difficult cases, such as "same objects with dissimilar appearances" and "different objects with similar appearances," are frequently present in bitemporal image pairs. Previous bitemporal fusion methods [19], [20], [21] usually subtract or concatenate two temporal features directly to obtain difference information. These naive operations are susceptible to varied lighting conditions. Therefore, it is essential to establish a meaningful connection between bitemporal images and develop fusion methods that can enhance adaptability and robustness for change detection. In addition, ground objects often exhibit complex geometrical structures and irregular boundaries, which can result in unsatisfying performance when detecting changes' boundaries.

Note that CNNs tend to mimic the bottom-up perception process of human vision [22], [23], and their global views of deep features lack spatial geometric information to determine local regions precisely. Consequently, the existing CNN methods [24], [25] often neglect the refinement of subtle changes from a top-down perspective, leading to inaccurate

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Fig. 1. Bitemporal image pairs of (a) "same objects with dissimilar appearances" and (b) "different objects with similar appearances" are shown. The red boxes in (a) indicate the unchanged area with large differences due to seasonal variation. The blue boxes in (b) indicate the changed area but the appearance is similar due to shadow.

boundaries of change regions, particularly within complicated scenes in high-resolution remote sensing images. Hence, it is conducive to carefully exploring the change details through a redefinition of low-level features under the guidance of high-level semantics, which aids in effectively learning fine changes.

Motivated by the above issues, we found that it remains to investigate how to aggregate bitemporal features for robust differences/similarities representation of changed/unchanged components and how to make low-level and high-level features work together for precisely characterizing change boundaries. On the one hand, the backbone encoder should extract representative paired features to activate the differences and similarities between the bitemporal images. We establish the correlation and aggregation of bitemporal features by combining feature difference and feature summation in multiscale, which aims to fuse bitemporal features to enhance changed and unchanged information. To mitigate the interference caused by lighting and shadows, we set a holistic attention module to adaptively fuse multilevel features with change-aware semantics to perceive the initial change map, boosting the perception of real changes. On the other hand, we modify the standard single decoder to a bifurcated one, which forms a cascaded dual decoder to delineate change boundaries in a coarse-to-fine manner. The dual-decoder architecture first generates an initial change map and subsequently utilizes it for further mask refinement. In the change location decoder, the initial coarse change map is yielded from the fused multilevel features to focus on the global change context. In the mask refinement decoder, the coarse change map acts as the guide, directing the low-level features toward the spatial details of the bitemporal changes, paying more attention to the refinement of change region boundaries.

Based on this, we propose a change-aware cascaded dual-decoder network (CACD2Net) that effectively extracts critical difference information and suppresses irrelevant background interference, enabling better detection and refinement of changes in complex scenes. The CACD2Net consists of a difference feature representation module (DFRM), a region perception module (RPM), a guidance attention module (GAM), and a difference enhancement module (DEM). In the feature extraction encoder, the DFRM with differential and additive operations is proposed to fuse the bifeatures at multiple levels to learn change-aware features. Within the cascaded dual-decoder framework, the first decoder's RPM applies the attention mechanisms to initially recognize the changes. Meanwhile, the second decoder utilizes GAM and DEM to strengthen low-level features that encompass texture and structural information, contributing to the fine-tuning of boundaries. The contributions of our work can be summarized as follows.

1) We propose the CACD2Net under the coarse-to-fine framework to learn representative bitemporal features for change detection. The network optimizes the utilization of distinct stage features and redefines their roles, enabling a hierarchical learning process from coarse-to-fine levels, which mitigates the impact of disturbing factors in complex scenes.
2) The proposed DFRM extracts difference information related to changes, which integrates the bitemporal features at multiple levels to excite the differences and similarities of changed and unchanged components. The highlighted information regarding feature differences is the foundation for subsequent network learning.
3) We propose the cascaded dual-decoder architecture for progressive change detection, where the RPM is devised to identify approximate change regions with contextual information, while the GAM and DEM are designed to focus on the injection of subtle changes aimed to further refine the change mask boundaries.
4) We conduct extensive experiments on two benchmark change detection datasets to evaluate the performance of the proposed CACD2Net comprehensively. Through quantitative and qualitative comparisons with other models, our method demonstrates superior advantages in detection results.

The rest of this article is organized as follows. Section II recalls the popular network framework and attention mechanism for change detection methods. Section III describes our proposed method in detail. Some experimental results are reported in Section IV. The conclusions are drawn in Section V.

II. Related Work

A. Network Structure for Change Detection

Deep neural networks can learn hierarchical, complicated, and nonlinear features from raw data automatically, allowing for better performance and robustness of models, so researchers now often use deep learning methods for change detection [26]. These methods can be roughly divided into two groups, early fusion and late fusion. The methods based on early fusion fuse bitemporal images into a single input and then feed it into neural networks to learn changes. Gong et al. [27] directly get a change detection map from two images using a trained deep neural network. Daudt et al. [19] exploited the UNet structure in

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change detection. However, this approach is not tailored to the specific characteristics of the change detection task, simplifying it to a semantic segmentation task and resulting in reduced detection accuracy. Nowadays, the more common approach is late fusion, which utilizes the Siamese network to extract bitemporal features and obtain the change detection result.

Daudt et al. [19] first applied the Siamese network [28] to remote sensing image change detection. The Siamese network is used to extract features from pre- and post-temporal images separately, and these features will be processed in a decoder to get results. In this way, the network can extract discriminative features and improve detection performance efficiently, so this two-branch network structure has become a popular framework for change detection tasks. The SNUNet proposed by Fang et al. [29] adopts the idea of dense connectivity based on UNet++ [30] and showed that aggregating multiscale features can improve the discrimination of change and nonchange regions. In order to model temporal relationships, Mou et al. [31] introduced long short-term memory (LSTM) networks to change detection tasks. Chen et al. [18] utilized the Transformer [32] module to enhance the features extracted from the final layer of the CNN backbone, enabling effective contextual modeling in the spatial–temporal domain. To address the discrepancy across different land-cover distributions in multitemporal images, Yang et al. [33] proposed asymmetric Siamese network (ASN) for semantic change detection. In addressing the challenges of building change detection with tilted viewing angles, Pang et al. [34] introduced three auxiliary tasks to effectively alleviate the mismatch of nearby buildings and the semantic ambiguity of building facades.

Although the multilevel features generated by deep neural networks can help to fully understand the input data, Han et al. [35] pointed out that there is a large amount of redundant information in these rich feature maps. In NAS-FPN [36], multilevel features are combined across levels. It is worth noting that models, such as Cascade RCNN [37] and Cascade DETR [38], have achieved notable success in object detection tasks by adopting cascade architecture to utilize multilevel features. Object detection is typically performed at the object level. However, applying the cascade scheme to pixel-level prediction tasks, such as change detection, is also a worthwhile consideration. Some studies [39], [40] in image semantic segmentation have confirmed that adopting a reoptimization strategy (i.e., coarse to fine), which involves multiple substeps, can achieve satisfactory performance. This inspires us to address the problems of change localization and boundary refinement separately. We decouple the change detection task into two substeps via a cascaded dual-decoder network architecture, allowing for the full utilization of bitemporal features at different stages, which aims at a more precise and detailed description of the changes, enhancing the overall detection performance.

B. Attention Mechanism

Recent studies have shown that incorporating attention mechanisms can greatly enhance the performance in various tasks, including image classification [41], image segmentation [42], and change detection [43]. Rather than simply adding or multiplying features, the principle of these attention mechanisms is to reweigh the features by calculating the attention weights for each position or channel, thus enabling the network to make better use of information. Attention mechanisms can usually be divided into two categories: channel attention and spatial attention.

The squeeze-and-excitation that is a channel attention module proposed by Hu et al. [44] can adaptively learn the interchannel relationships of features through squeezing and stimulation operations, thus improving the learning ability of neural networks. In order to make the network focus on important channels or spatial locations, some approaches incorporate both channel attention and spatial attention. For example, the convolutional block attention module (CBAM) [45] achieves cooperation between spatial and channel attention through sequential cascades. This integrated use of attention can be better adapted to feature extraction in different tasks. Based on this, Zhang et al. [46] introduced CBAM into the change detection task to effectively fuse features in different domains. Cheng et al. [47] used channel attention in the encoder to facilitate semantic-specific feature extraction and spatial attention in the decoder to enhance feature refinement. They combined spatial attention and channel attention to get higher change detection accuracy. Another example is nonlocal modules [48], which can capture long-range dependencies on feature maps and improve the feature representation of the network. Chen et al. [49] introduced a dual-attention module to capture long-range dependencies, which enhanced the network's ability to discriminate between changed and unchanged regions effectively. Lei et al. [50] proposed a remote sensing image change detection method based on a spatial–spectral nonlocal (SSN) strategy. Although these methods enable the attention mechanism to adaptively adjust weights to focus on changes, the fact that they need to learn these changes from scratch in an unconstraint way may lead to false detection. Our design utilizes the coarse map obtained from the first decoder as prior information for change perception, enabling the second decoder to pay more attention to the boundary details of real changes.

III. Methodology

A. Overview

The framework of our proposed CACD2Net is shown in Fig. 2. It is composed of three phases: feature extraction (backbone encoder), change location, and mask refinement. The latter two components form a cascaded dual-decoder architecture. Given the imaging condition variations and complex boundaries of change regions in high-resolution remote sensing images, we adopt a coarse-to-fine strategy to detect changes. The "coarse" part is responsible for detecting the approximate locations of changes, while the "fine" part leverages the prior information from the former decoder to further refine pixelwise boundaries. In the following, we introduce in detail the designed modules in our framework: 1) DFRM, which fuses bitemporal features to extract change representation;

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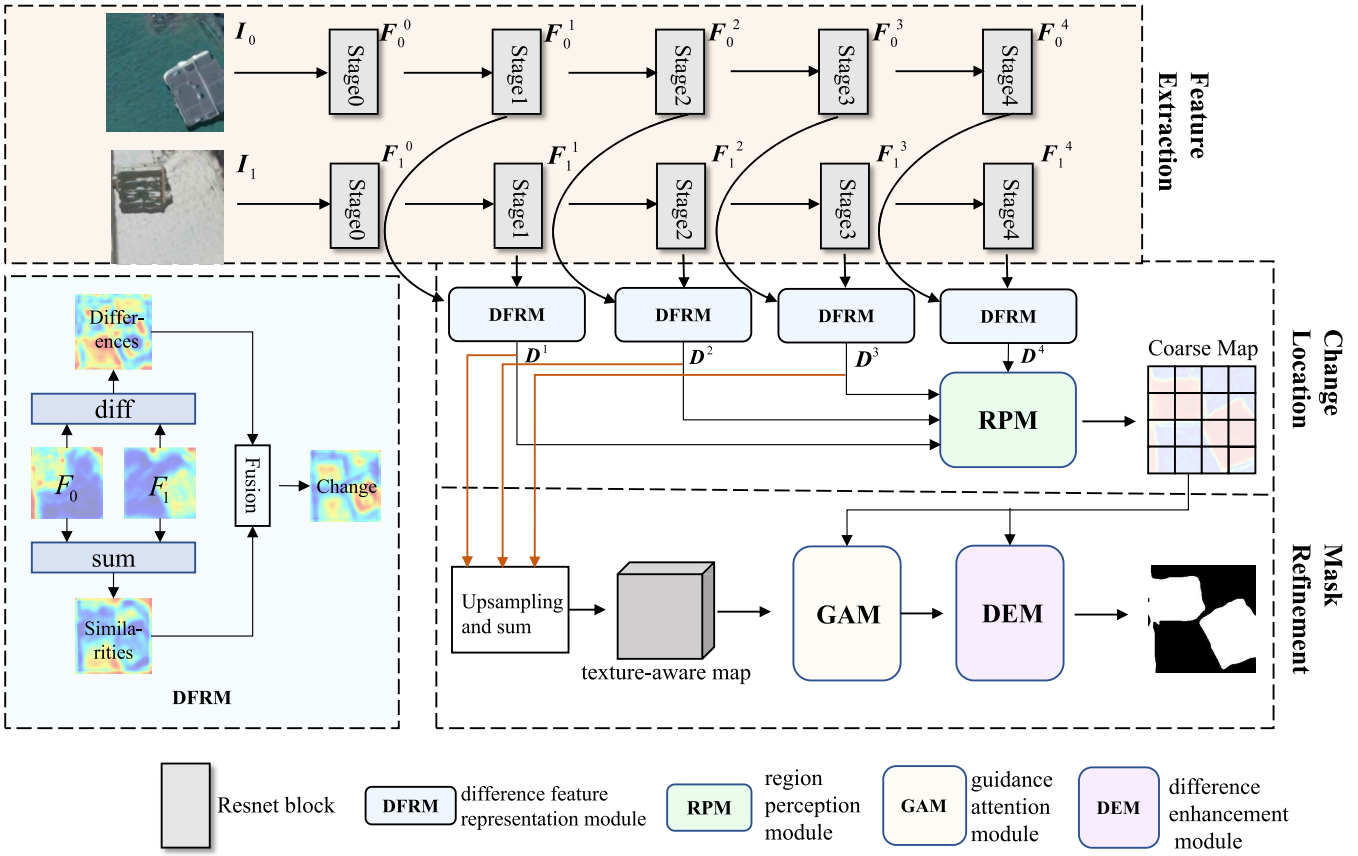


Fig. 2. Structure of the proposed CACD2Net. The network can be divided into three parts: feature extraction, change location, and mask refinement. The DFRM extracts the difference features for each stage, the RPM generates the coarse change map to locate rough change region, and the GAM, as well as the DEM, refines the change boundaries to obtain the final intricate change map.

2) RPM, which is designed to locate the change regions; 3) GAM; and 4) DEM, which are used to refine the change mask boundaries.

B. Feature Extraction

A pair of images captured at different times are input into the network and are denoted by $\mathbf{I}_0$ and $\mathbf{I}_1$. A set of bitemporal features $\mathbf{F}_0^i$ and $\mathbf{F}_1^i$ $(i \in 0, 1, 2, 3, 4)$ are extracted from ResNet34 [51] backbone, respectively, where i indexes the feature extraction stage in the ResNet34. By processing the input data with multiple convolutional layers, these features can capture relevant patterns, textures, and structures for each of the bitemporal images. Note that the original ResNet34 has five stages, each with downsampling by 2. We make a change by setting the stride of the last two stages to 1 instead of 2 to reduce the loss of spatial details.

C. Change Location

1) Difference Feature Representation Module: In change detection tasks, the fusion of bitemporal features is typically a crucial component. Many methods [25], [52], [53] directly use the operations of subtraction or concatenation to fuse bitemporal features to obtain the difference feature. However, such naive manipulations make it difficult to extract robust feature representation due to varied lighting and shadows in bitemporal remote sensing images. In order to address this problem, we propose the DFRM to enhance the discriminative features of the concerned change components and suppress the undesired interference.

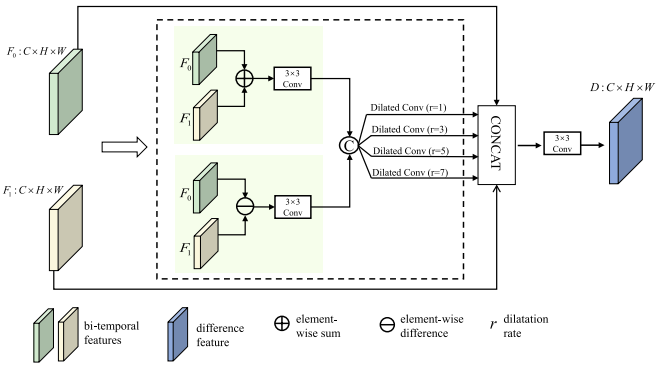


Fig. 3. Illustration of our DFRM.

The structure of DFRM is shown in Fig. 3. For bitemporal features $\mathbf{F}_0^i, \mathbf{F}_1^i \in \mathbb{R}^{C^i \times H^i \times W^i}$, where W, H, and C denote the width, height, and channel number, we first process them by applying the addition and subtraction operations. After the 3×3 convolution blocks, we fuse the processed two features to extract comprehensive feature representation. We set up four parallel branches with different dilation rates $r \in \{1, 3, 5, 7\}$ to expand the receptive field to capture the changes at different scales. In the end, the output of the four branches, and $\mathbf{F}_0^i$ and $\mathbf{F}_1^i$ are concatenated. The original features are reused

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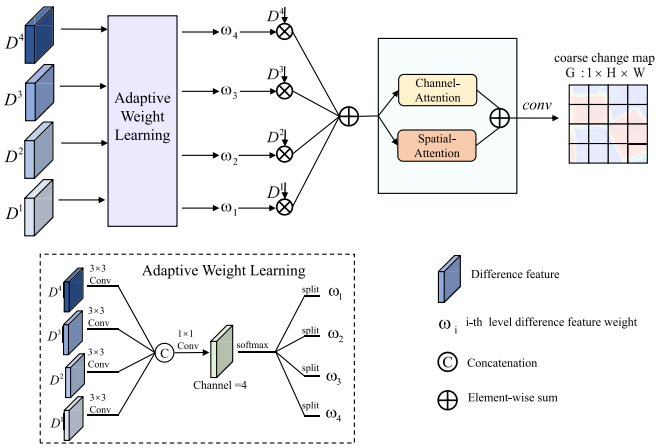


Fig. 4. Illustration of our RPM.

to complement the missing details. It is worth emphasizing that each convolution block consists of a convolutional layer, a batch normalization layer, and an activation layer. As shown in Fig. 2, the DFRM is embedded in each stage of the ResNet34, and the generated difference features from DFRM are denoted by $\mathbf{D}^i (i \in 1, 2, 3, 4)$. To simplify the utilization of the difference features at each level in the two decoders, we uniformly set the channel number of the DFRM output features to 64.

In our design, the subtraction and addition operations are used to focus on the differences and similarities of changed and unchanged components, respectively. The subtraction operation can distinguish the change components from their surroundings. We excite the regions where changes occurred by calculating the differences between bitemporal features. In contrast, if the feature responses at the same spatial position in two feature maps are very similar, then by addition operation, we can obtain the cumulative effect of similarities to facilitate the capture of consistency for unchanged components. When dealing with unchanged components, which tend to occupy large regions in the bitemporal images, dilated convolution utilizing a larger receptive field will better identify these regions. In contrast, when handling unchanged components, as the changed regions are typically smaller and involve local differences, operating within a smaller receptive field allows the model to better capture these local details.

2) Region Perception Module: The RPM module in the network provides initial change map, which is further optimized by next decoder. The difference features $\mathbf{D}^i (i \in 1, 2, 3, 4)$ are dynamically fused with adaptive weights to accommodate changes at multiscale. Meanwhile, we use the attention mechanisms to enhance the semantic information of the changes. The structure of the proposed RPM is shown in Fig. 4.

The multilevel difference features can capture changes at different scales. Inspired by the adaptive spatial feature fusion [54], we introduce a dynamic weight allocation mechanism for adaptive feature fusion in RPM, assigning learnable weights to the difference features at each layer. High-level features, with a larger receptive field, can capture more extensive contextual information in the image, which is crucial for detecting large-scale changes. Meanwhile, low-level features focus more on details and local information, making them more sensitive to small-scale changes. By assigning different weights to these features and merging them, the model becomes more flexible in adapting to various scales and shapes of changes. By assigning different weights to these features and merging them, the model becomes more flexible in adapting to various scales and shapes of changes. Specifically, for the adaptive weight learning in Fig. 4, we apply a 3 × 3 convolution to each difference feature $\mathbf{D}^i$, which are then concatenated along the channel dimension. Next, a 1 × 1 convolution operates on this fused feature, generating a tensor with four channels. Each channel of the tensor ω_i represents the weight corresponding to the ith difference feature. The four features are then elementwise multiplied by their respective weights and subsequently summed. We adopt channel and spatial attention mechanisms to further enhance the representative power of the fused feature. The attention mechanisms help the model focus more on the overall context when inferring the change regions. The enhanced feature is processed by two 3 × 3 convolution blocks, and we obtained the one-channel coarse change map $\mathbf{G} \in \mathbb{R}^{1\times H\times W}$, which will be involved in the loss function and guide the subsequent refinement of the change mask.

D. Mask Refinement

1) Guidance Attention Module: As discussed in Section III-C, our coarse change map $\mathbf{G} \in \mathbb{R}^{1\times H\times W}$ can capture a relatively rough location of the changes, but its structural and textural details need to be further refined. In the mask refinement phase, we aim to enhance the delineation of change boundaries. By utilizing the subtle change information in the low-level features, we can better capture and highlight the boundaries of the detected changes. Due to the well-balanced integration of semantic and spatial information, our primary focus for mask refinement is based on the low-level difference features.

The GAM can be seen in Fig. 5, where the coarse map $\mathbf{G}$ will guide low-level features for mask refinement. To compensate for the loss of structural and textural information, we leverage low-level difference features of the first three layers, i.e., $\mathbf{D}^1$, $\mathbf{D}^2$, and $\mathbf{D}^3$. These features are upsampled to the same size (standardized on $\mathbf{D}^1$ size) and then fused through elementwise addition, aiming to minimize further loss of detailed information. We refer to the fused feature as a texture-aware map $\mathbf{T} \in \mathbb{R}^{C\times H\times W}$. In addition, $\mathbf{G}$ is copied along the channel dimension. After that, it is reshaped and transposed to $\mathbf{K} \in \mathbb{R}^{HW\times C}$, while $\mathbf{T}$ is reshaped to obtain $\mathbf{Q} \in \mathbb{R}^{C\times HW}$, and matrix multiplication is performed between $\mathbf{Q}$ and $\mathbf{K}$. The obtained resulting feature is $\mathbf{R} \in \mathbb{R}^{C\times C}$, which is softmax-normalized afterward. The processing of $\mathbf{R}$ can be represented by the following formula:

$$R_i = \frac{\exp(Q_i \times K_i)}{\sum_{j=1}^{C} \exp(Q_j \times K_j)}. \tag{1}$$

$\mathbf{R}$ is multiplied with $\mathbf{Q}$ and then reshaped to form the attention map $\mathbf{A} \in \mathbb{R}^{C\times H\times W}$, which focuses on the regions in $\mathbf{T}$ where changes occur. Finally, the attention map is formed into a residual connection with $\mathbf{T}$ to obtain the map $\mathbf{T}_{\text{new}}$.

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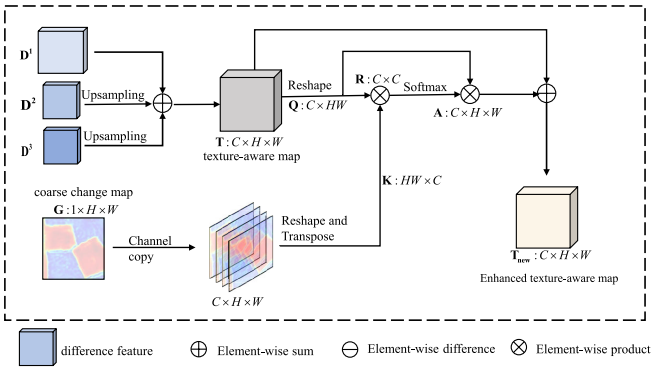


Fig. 5. Illustration of our GAM.

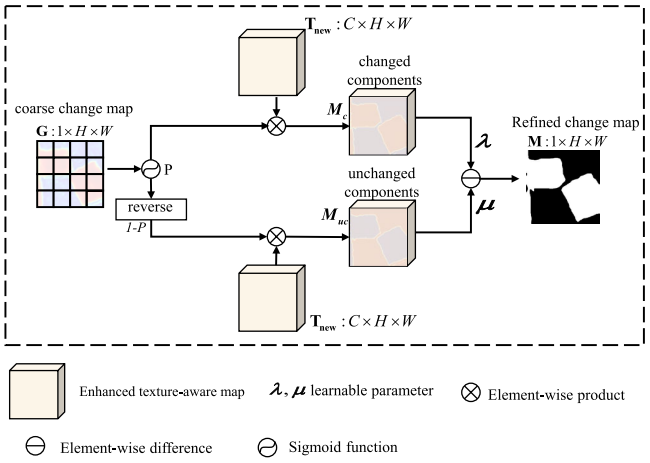


Fig. 6. Illustration of our DEM.

In this process, texture-aware map **T** complements the lost detailed information in the coarse change map. The map **G** that provides the prior knowledge plays a guiding role in directing the learning of the map **T** toward the precise change location. This integration of information enhances the capability of the map **T** to effectively identify and localize changes within the bitemporal images.

2) Difference Enhancement Module: This proposed module is used to enhance the differences between the change regions and surrounding regions to further distinguish them. When performing change detection, many existing methods attempt to detect both change and nonchange components jointly, and they fail to emphasize the semantic difference between the change regions and their surrounding regions. Our method implicitly separates the change foreground and the nonchange background to learn the features specifically. Then, we adaptively fuse the two features to reinforce the differences between them. Through the previous steps of coarse localization and adjustment of the change regions, the change and nonchange regions have been basically delineated. By further adopting the DEM of separate processing and learning, we can gain a better understanding and description of the intrinsic features of change and nonchange components, thereby mitigating the mutual interference between the two components in complex scenes.

As shown in Fig. 6, the coarse change map **G** applies the sigmoid function to get the map $\mathbf{P} \in \mathbb{R}^{1\times H\times W}$, where each element value ranges from 0 to 1. In the **P**, the activation values of the changed regions tend to be 1, indicating the presence of change, while the unchanged regions tend to have a value of 0. Conversely, map $1-\mathbf{P}$ represents the opposite. Thus, **P** is used to highlight the changed regions in $\mathbf{T}_{\text{new}}$, while $1-\mathbf{P}$ is used to enhance the unchanged regions.

To expand the semantic distinguishability between the "change" and "unchange," the two probability maps **P** and $1-\mathbf{P}$ are multiplied with $\mathbf{T}_{\text{new}}$, controlled by the corresponding learnable parameters λ and μ. The enhanced results of the "change" and "unchange" are denoted as $\mathbf{M}_{\text{c}}$ and $\mathbf{M}_{\text{uc}}$, respectively, which are combined through differential operation to produce the final refined change detection map $\mathbf{M} \in \mathbb{R}^{1\times H\times W}$. The process can be expressed by the following formula:

$$\begin{aligned}\mathbf{M} &= \lambda\mathbf{M}_{\text{c}} - \mu\mathbf{M}_{\text{uc}} \\ &= \lambda\mathbf{P}\mathbf{T}_{\text{new}} - \mu(1-\mathbf{P})\mathbf{T}_{\text{new}}.\end{aligned} \tag{2}$$

E. Network Details

1) Backbone: As mentioned earlier, we use a modified ResNet34 to extract bitemporal features. The original ResNet34 has five stages, each with downsampling by 2. We change the stride size of the last two stages to 1 to maintain the resolution.

2) Loss Function: Usually, in binary change detection, there are much fewer positive samples than negative samples. This imbalance problem always invalidates the binary cross-entropy (BCE) loss. Since BCE loss has identical weights for positive and negative samples, the gradient of negative samples always dominates the training process. To alleviate this problem, we choose the weighted BCE loss as the loss function for coarse change map **G** and refined change map **M**. Thus, the total loss is calculated as follows:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_i\left[y_i \log \hat{y}_i + (1-y_i)\log\left(1-\hat{y}_i\right)\right] \tag{3}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_G + \mathcal{L}_M \tag{4}$$

where N is the number of training samples, y_i stands for the ith sample's label, $\hat{y}_i$ indicates the predicted value of the ith sample, and w_i denotes the ith sample's weight parameter in loss.

3) Inference of CACD2Net: Upon optimizing CACD2Net, the trained network can be utilized to predict changes between bitemporal images. The inference details of our CACD2Net are presented in Algorithm 1, illustrating the workflow of the proposed method.

IV. Experiments

A. Datasets

1) SYSU-Change Detection Dataset [55]: The Sun Yat-sen University change detection (SYSU-CD) is a challenging dataset that provides a wide range of change types in relatively complex scenes. The dataset collected 20 000 pairs of 0.5-m aerial images, each of size 256 × 256, taken in Hong Kong between 2007 and 2014. The 12 000/4000/4000

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Algorithm 1 Inference of CACD2Net for Change Detection

Input: $I = (I_0, I_1)$ (a pair of registered bitemporal images)
Output: M (a prediction change mask)
Step1: extract image features at each level by CNN backbone;
for $i \leftarrow 0$ **to** 1 **do**
  for $j \leftarrow 1$ **to** 4 **do**
    $F_i^j = CNN_Backbone(I_i)$;
  end
end
Step2: use DFRM to extract difference information related to changes at multiple levels;
for $j \leftarrow 1$ **to** 4 **do**
  $D^j = DFRM(F_0^j, F_1^j)$
end
Step3: use change location decoder to identify approximate change regions;
$G = change_location_decoder(D^1, D^2, D^3, D^4)$
Step4: use mask refinement decoder to refine the change mask;
$M = mask_refinement_decoder(D^1, D^2, D^3, G)$

pairs of these labeled bitemporal images are used for training/validation/testing, respectively.

2) SVCD Dataset [56]: Season-varying change detection (SVCD) is a commonly used dataset in remote sensing image change detection. It consists of 16 000 pairs of bitemporal RGB remote sensing images taken in different seasons, each with a size of 256 × 256 pixels, and their corresponding labels. Among them, 10 000 pairs are used for the training, while 3000/3000 pairs are used for the validation/testing.

B. Implementation Details

We use the Python programming language and the PyTorch library to implement the proposed change detection method. The initial learning rate is set to 0.003, and batch size is set to 16. The number of epochs is set to 100 for the SYSU-CD dataset and 300 for the SVCD dataset. The gradient descent optimization method is stochastic gradient descent (SGD) with Nesterov momentum. The augmentation strategies include horizontal flipping, vertical flipping, cropping, and blurring. All experiments in this work were performed on a workstation equipped with an NVIDIA 3090 GPU.

C. Evaluation Metrics

In this article, we adopt overall accuracy (OA), intersection over union (IoU), and F1 scores as evaluation metrics. The latter two metrics are typically used to quantify the performance of change detection, and note that they represent only the metrics of the change category. The values of IoU and F1 range from 0 to 1, and the higher the value, the better the performance. The metrics are calculated as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \tag{7}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{8}$$

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{9}$$

where TP, FP, TN, and FN represent true positive, false positive, true negative, and false negative, respectively.

D. Comparison With State of the Art

1) *FC-EF [19]:* The bitemporal images are concatenated along channel dimension as a single input, and the difference features are extracted by a fully convolutional network (FCN).
2) *FC-Siam-Diff [19]:* Siamese FCN is used to extract bitemporal features separately, and change information is extracted by feature subtraction.
3) *FC-Siam-Conc [19]:* Siamese FCN is used to extract bitemporal features separately, and change information is extracted by feature concatenation.
4) *IFN [46]:* The method applies channel and spatial attention to decode bitemporal features at all levels, whose supervised loss of each level in the decoder is employed to train the intermediate layers better.
5) *SNUNet [29]:* Dense skip connections between encoders and decoders are used to maintain a high-resolution, fine-grained feature representation.
6) *BIT [18]:* Transformer-based change detection method. The bitemporal images are represented as several tokens. Then, through the self-attention mechanism, the model is able to learn the correlations between different regions in the images.
7) *ISNet [47]:* This method employs edge maximization strategies to enhance the gap between change and unchange semantics and also employs attention mechanisms to highlight both semantic and spatial information.

To evaluate the effectiveness of our proposed CACD2Net, we conducted extensive experiments on the two change detection datasets to obtain quantitative results. The visualization of comparison results of different models on the SYSU-CD and SVCD test sets is given, respectively. Our method achieves the best overall performance and robustness on both datasets. This verifies that our proposed CACD2Net is valuable and constructive for the change detection task.

1) Comparison on SYSU-CD Dataset: As shown in Table I for the quantitative results of the SYSU-CD dataset, FC-conc has higher precision and low recall, while our method achieves the best performance for F1, OA, and IoU. For visualization comparison, Fig. 7 shows the examples of the results predicted by different methods on the SYSU-CD dataset. Most of the methods are poor at detecting boundaries (rows 1 and 2), while our approach generates the change maps, which have

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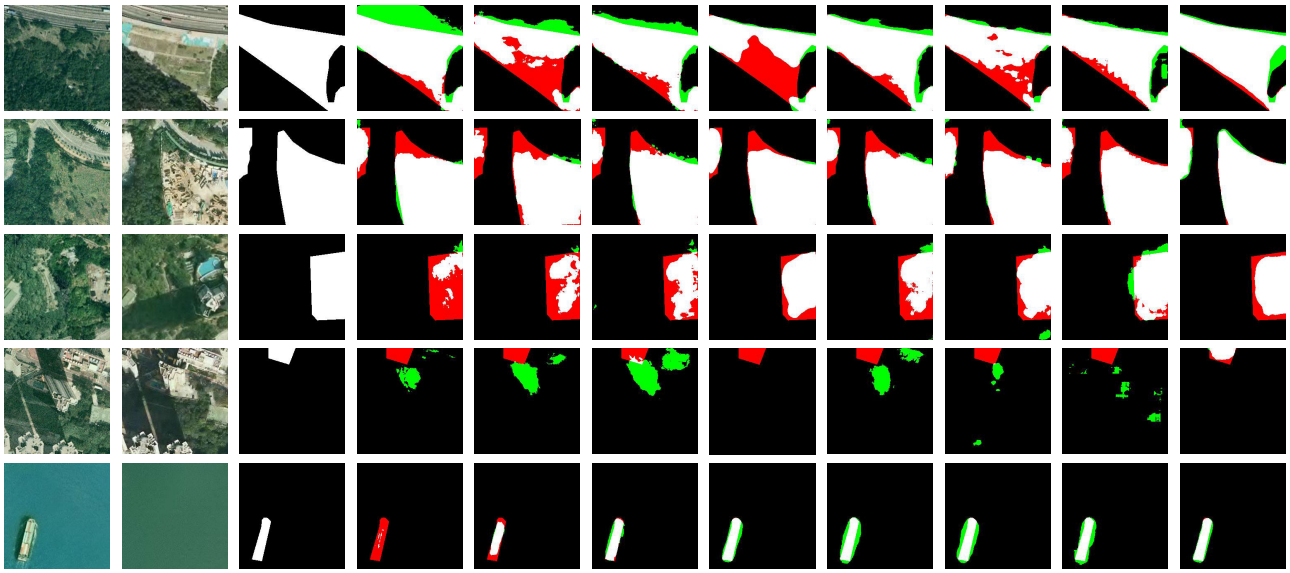


Fig. 7. Performance of different methods on SYSU-CD dataset. From left to right: (a) T1 instance, (b) T2 instance, (c) GT, (d) FC-EF, (e) FC-Diff, (f) FC-Conc, (g) IFN, (h) SNUnet, (i) BIT, (j) ISNet, and (k) our method. Colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), green for FP, and red for FN.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SYSU-CD DATASET. THE BEST RESULTS HAVE BEEN **BOLDED**

Methods	precision	recall	OA	IoU	F1
FC-EF	80.31	67.04	88.35	57.58	73.08
FC-Diff	81.33	66.40	88.48	57.62	73.11
FC-Conc	**90.18**	48.28	86.56	45.87	62.89
IFN	82.39	73.57	90.06	63.57	77.73
SNUNet	78.41	73.38	88.96	61.05	75.81
BIT	79.18	77.01	89.80	64.04	78.08
ISNet	80.27	76.41	90.01	64.44	78.29
CACD2Net (ours)	84.65	**83.02**	**92.45**	**72.16**	**83.83**

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SVCD DATASET. THE BEST RESULTS HAVE BEEN **BOLDED**

Methods	precision	recall	OA	IoU	F1
FC-EF	83.92	56.31	93.57	50.83	67.39
FC-Diff	79.71	59.56	93.44	51.72	68.18
FC-Conc	75.94	62.33	93.22	52.06	68.47
IFN	**97.15**	91.70	98.70	89.29	94.34
SNUNet	94.82	92.45	98.51	88.00	93.62
BIT	95.31	87.31	98.00	83.71	91.13
ISNet	95.18	94.43	98.78	90.12	94.80
CACD2Net (ours)	94.10	**98.11**	**99.05**	**92.42**	**96.06**

smoother and complete boundaries. The change results of our net also maintain high internal compactness (row 3). As shown in row 4, while most methods are prone to generating false positives when detecting small changes in shadows, our method still shows good results. As shown in row 5, our method is also better at identifying the shape of objects. From the above results, it can be seen that our method generates finer boundaries while also maintaining superior detection performance in the presence of shadow occlusion.

2) Comparison on SVCD Dataset: According to Table II, our method achieves the best results compared with other networks. The visual analysis of the experimental results for the SVCD dataset is shown in Fig. 8. Small and sparse changes make the detection results more susceptible to noise, leading to some inaccuracies in the detections (row 1), while our method is better equipped to address this situation more effectively. The results in rows 2 and 3 show the superiority of our method for detecting changes at different scales. When detecting narrow road changes (rows 4 and 5), FC-EF, FC-Conc, FC-Diff, and IFN barely detect the changes, while BIT and ISNet have some missing areas. Compared with other models, our network is more accurate in dealing with the irregular boundaries of change regions.

The better performance is primarily attributed to our "coarse-to-fine" change detection scheme. Instead of directly fusing multilayer features or relying solely on the last layer's features, our approach selectively utilizes the distinct characteristics of each layer of features to generate change maps. It was also found that our method could detect changes at different scales, because we designed the network to use not only dilated convolution in the module but also nonlocal units by attention mechanism. When objects undergo changes in complex environments (e.g., light variation and shadow occlusion), false detections and missed detections are likely to occur due to the high noises. Our network still achieves good performance in terms of the integrity and stability of the detected changed objects.

E. Ablation Analysis

In this section, we conducted comprehensive ablation experiments on the SYSU-CD dataset. In our ablation experiments, we build the baseline of the single-decoder change detection network as one that includes only the backbone network and a classifier (consisting of two convolution layers), and the way of bitemporal feature fusion is $|x1 - x2|$. The dual-decoder network is an extension of the single-decoder baseline, incorporating either GAM or DEM. To further

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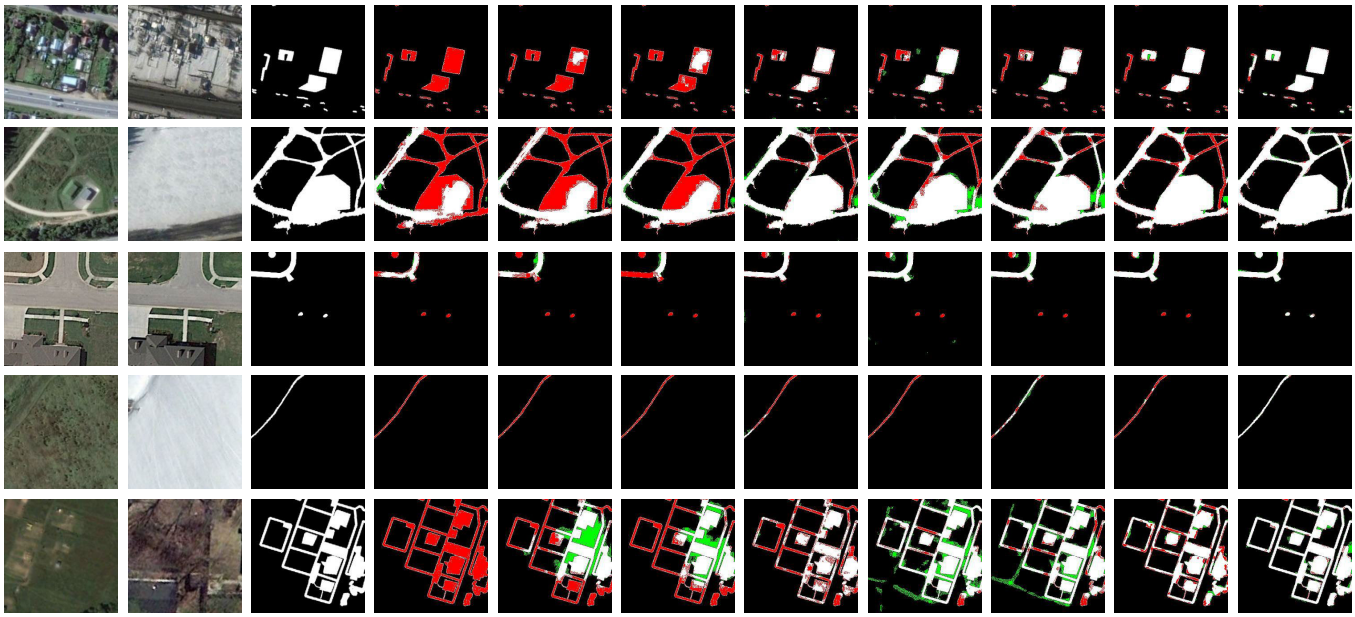


Fig. 8. Performance of different methods on SVCD dataset. From left to right: (a) T1 instance, (b) T2 instance, (c) GT, (d) FC-EF, (e) FC-Diff, (f) FC-Conc, (g) IFN, (h) SNUnet, (i) BIT, (j) ISNet, and (k) our method. Colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), green for FP, and red for FN.

TABLE III
ABLATION STUDIES ON THE PROPOSED MODULES

Methods	DFRM	RPM	GAM	DEM	OA	IoU	F1
single-decoder					89.78	64.10	78.12
	✓				90.48	67.21	80.38
		✓			90.81	67.86	80.85
	✓	✓			91.36	69.24	81.82
dual-decoder			✓		91.10	68.67	81.42
				✓	90.55	67.90	80.89
			✓	✓	91.58	69.68	82.12
	✓	✓	✓	✓	**92.44**	**72.16**	**83.83**

validate the effectiveness of our method, we incrementally add the proposed modules to "single decoder" to evaluate the performance of our network architecture.

1) Quantitative Analysis: The experimental outcomes detailed in Table III illustrate the impact of each module on the network's performance. In the "single-decoder" baseline, the introduction of the DFRM makes the network more sensitive to real changes, resulting in 2.26% increase on F1 score. By incorporating the RPM, the semantic information related to changes is reinforced, leading to 2.73% improvement on F1 score. The combination of the DFRM and the RPM further strengthened the network's ability to perceive changes, increasing the F1 score by 3.7%. After introducing the dual-decoder architecture, the GAM and the DEM, respectively, contribute to an increase of 3.3% and 2.77%. The combined utilization of GAM and DEM refines the delineation of change boundaries more accurately, resulting in 4% improvement compared with the baseline. The integration of all four modules (our CACD2Net) significantly surpasses the "single-decoder" baseline by 5.71%.

2) Visualizations of Ablation Experiments: We provided some visualizations of ablation experiments, as shown in Fig. 9, to better illustrate the contributions of each module. As can be seen from (d)–(g), when ablation experiments are performed on the DFRM and RPM with a single decoder, both modules play a good role in the localization of the change regions, and localization accuracy is further enhanced when they are used together. From (h)–(k), then, it can be seen that GAM and DEM can make the change region more precise and clearer, which proves that our designed dual-decoder structure improves the boundary delineation.

3) Discussion on DFRM: We design a new fusion module to reduce pseudo-changes, and we compare other common fusion methods to illustrate the effectiveness of our module. As shown in Table IV, we compare the proposed DFRM with $|x1 - x2|$, $x1 + x2$, and concat$(x1, x2)$, commonly used in change detection for the fusion of bitemporal features. We can see that our module has a very significant effect, which benefits subsequent feature understanding. We noticed that DFRM has a slight decrease in precision compared with the conventional operation concat$(x1, x2)$. The design of the DFRM module aims to capture all possible changes, thereby reducing missed detections and ensuring a relatively higher recall rate. Subsequent modules in the cascaded architecture are able to further rectify false positives, thus achieving better overall performance. In the DFRM, the subtraction operation is beneficial to highlight dissimilar features, and the addition operation is beneficial to strengthen similar features, and the combination of the two can obtain more comprehensive information on changed and unchanged components. Moreover, the dilated convolution helps to expand the receptive field to explore the changes at different scales. The residual connection of the original bitemporal features helps to compensate for the lost information. DFRM is used to extract the difference features of different stages to provide change-aware representation.

4) Discussion on DEM: The DEM is designed based on the concept of separating foreground and background, where the

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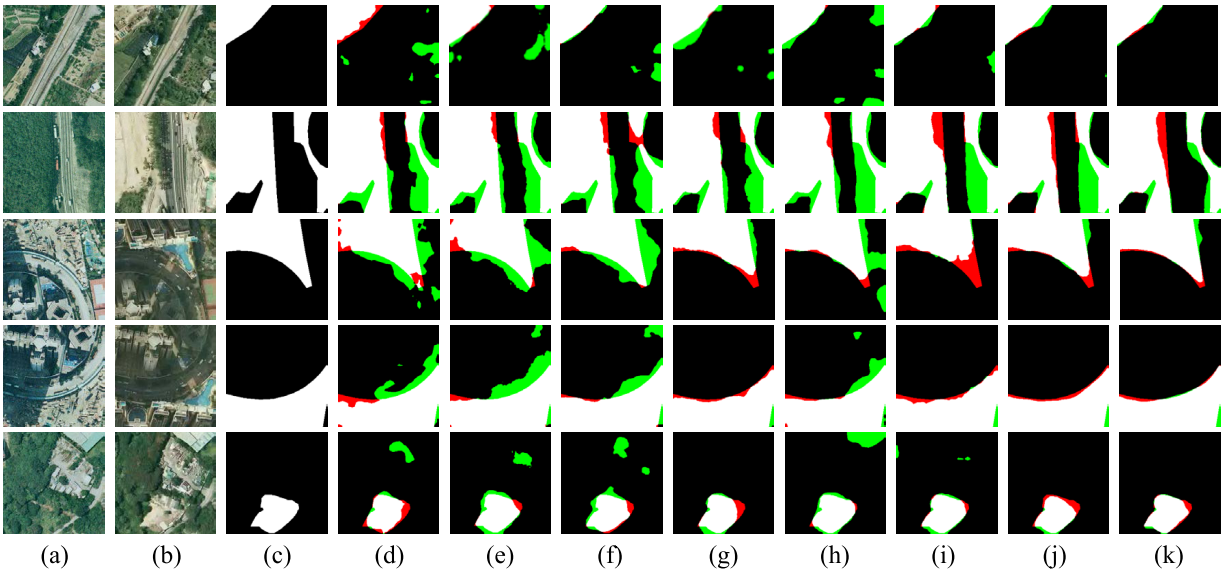


Fig. 9. Comparisons of different ablated models on SYSU-CD dataset. From left to right: (a) T1 instance, (b) T2 instance, (c) GT, (d) single decoder, (e) single decoder with RPM, (f) single decoder with DFRM, (g) single decoder with RPM and DFRM, (h) dual decoder with GAM, (i) dual decoder with DEM, (j) dual decoder with GAM and DEM, and (k) our CACD2Net. Colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), green for FP, and red for FN.

TABLE IV
EFFECT OF DIFFERENT FUSION METHODS

Fusion	precision	recall	OA	IoU	F1
$x1 + x2$	83.10	78.87	91.06	67.97	80.93
$\|x1 - x2\|$	84.02	81.35	91.95	70.45	82.66
$concat(x1, x2)$	**85.92**	79.92	92.18	70.67	82.81
DFRM (proposed)	84.65	**83.02**	**92.45**	**72.16**	**83.83**

TABLE V
ABLATION STUDIES FOR THE DEM

Methods	precision	recall	OA	IoU	F1
None	82.26	81.95	91.58	69.66	82.11
only $\mathbf{M}_c$	82.96	82.94	91.96	70.87	82.95
w/o λ, μ	80.99	**85.53**	91.85	71.22	83.19
$\mathbf{M}_c$-$\mathbf{M}_{uc}$ (proposed)	**84.65**	83.02	**92.45**	**72.16**	**83.83**

"change" regions are considered as the foreground, and the "nonchange" regions are regarded as the background. We try to enlarge the semantic difference between "change" and "nonchange" as much as possible, so in Table V, we present four sets of experimental results: the results without difference enhancement, the results with only foreground $\mathbf{M}_c$ enhancement, the results without learnable hyperparameters λ and μ, and the results with difference enhancement (i.e., $\mathbf{M}_c - \mathbf{M}_{uc}$). Table V clearly shows the effectiveness of the last one that is proposed. By separating the change regions from their adjacent unchanged backgrounds, we can obtain more specific difference features. Introducing learnable parameters equips the model with greater flexibility and adaptability. Since the model becomes more selective in borderline cases or difficult-to-identify changes, there is a slight decline of recall. Given the increased F1 score, the introduction of hyperparameters is necessary. It can be concluded that our module can effectively enhance the discriminability of changed targets from the surrounding areas, leading to the enhancement of difference features.

TABLE VI
COMPARISON WITH OTHER METHODS ON COMPUTATIONAL COMPLEXITY

Methods	FLOPs(G)	Params.(M)
FC-EF	3.57	1.35
FC-Diff	4.72	1.35
FC-Conc	5.32	1.55
IFN	82.26	50.44
SNUNet	54.83	12.03
BIT	7.66	3.04
ISNet	15.19	22.75
CACD2Net (ours)	53.72	28.75

F. Computational Complexity

Table VI exhibits the number of model parameters (Params.) and the floating-point operations per second (FLOPs), for which the size of input images is 256 × 256 × 3. Since our CACD2Net adopts the dual-decoder structure, although there is some trade-off in resource requirements, the detection performance has been significantly improved. Compared with IFN, though it employs the single-decoder architecture, our model featuring a dual-decoder design obtains far superior detection results with lower complexity. While there is still room for reducing model complexity, the dual-decoder design does steadily contribute to improvements in results.

V. CONCLUSION

In this article, we propose a CACD2Net for change detection. The network extracts difference features at multiple levels and progressively detects changes from coarse to fine using a cascaded dual-decoder architecture. Specifically, our proposed DFRM mitigates the impact of irrelevant information of varied imaging conditions. RPM facilitates adaptive feature correlation through attention mechanisms to roughly locate the change regions. GAM and DEM are responsible for further refining the boundaries of the changes. By focusing

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on exploring the discriminability between change regions and their surrounding regions, our method achieves better smoothness of boundaries and integrity of the internal areas. Our CACD2Net achieves state-of-the-art performance on two public datasets, and we validate the effectiveness of each module. In future research, we can further explore the generalization of the proposed method in urban-scale or large-scale change detection [57], accommodated to different scenarios for better applicability.

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